

Faculty of Engineering and Technology (FET)

Sri Sri University

Cuttack, Odisha, India

Pin-75400

B.Tech Project Report

Project Title: Bank Network

Group Number: 2

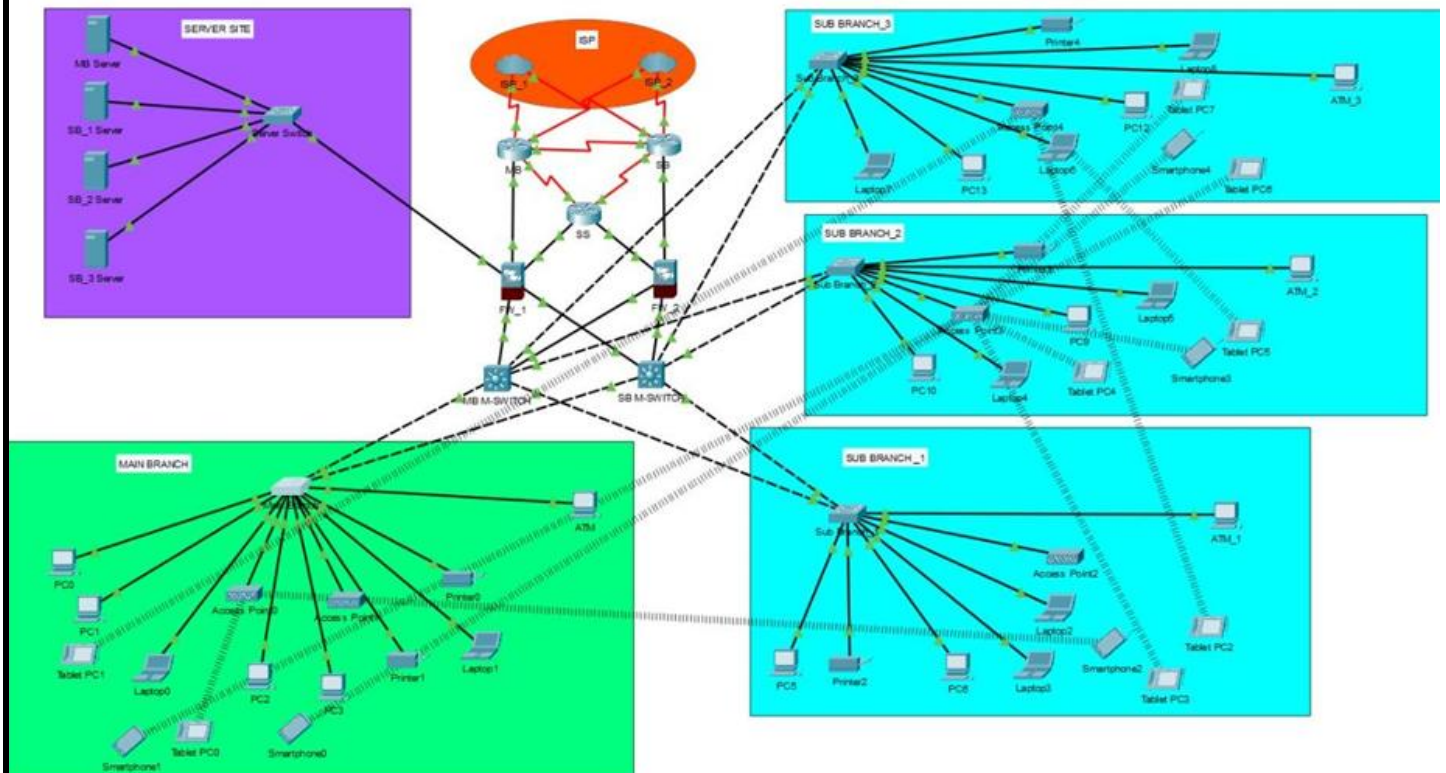
Department:

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SYSTEM DESIGN



VLAN Segmentation

```
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name SUB_BRANCH_1
Switch(config-vlan)#exit
Switch(config)#vlan 11
Switch(config-vlan)#name SUB_BRANCH_2
Switch(config-vlan)#exit
Switch(config)#vlan 12
Switch(config-vlan)#name SUB_BRANCH_3
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name MAIN_BRANCH
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name SERVER_ROOM
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name DMZ
Switch(config-vlan)#exit
Switch(config)#|
```

Firewall configuration

```
en
conf t
hostname FW_1

interface GigabitEthernet0/0
 nameif outside
 security-level 0
 ip address 100.1.1.2 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 nameif inside
 security-level 100
 ip address 192.168.20.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/2
 nameif server
 security-level 90
 ip address 192.168.30.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/3
 nameif to_FW2
 security-level 50
 ip address 192.168.50.1 255.255.255.0
 no shutdown

route outside 0.0.0.0 0.0.0.0 100.1.1.1

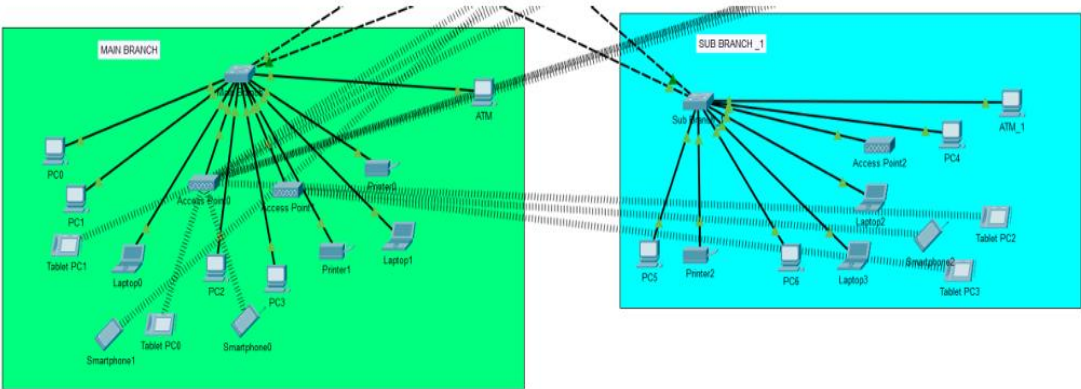
access-list INSIDE_ACL extended permit ip any any
access-list SERVER_ACL extended permit ip any any

access-group INSIDE_ACL in interface inside
access-group SERVER_ACL in interface server

nat (inside,outside) dynamic interface
nat (server,outside) dynamic interface

exit
```

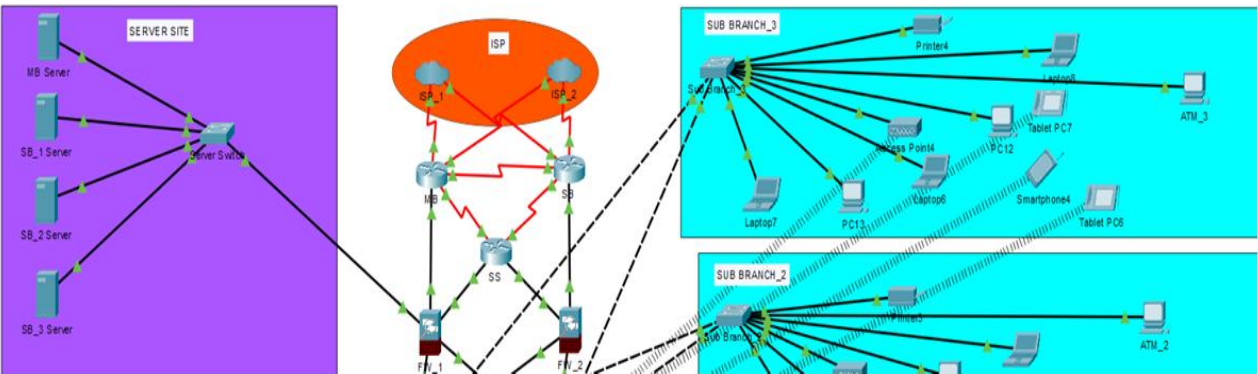
2.MAIN BRANCH <-> SUB BRANCH



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC5	ICMP		0.000	N	0	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC5	PC0	ICMP		0.000	N	0	(edit)	(delete)

5.SUB BRANCH <-> SERVER SITE



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC13	SB_3 Server	ICMP		0.000	N	0	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	SB_3...	PC13	ICMP		0.000	N	0	(edit)	(delete)

REFERENCES

- Cele, N.N. and Kwenda, S. (2025) Identified identity theft, malware attacks, phishing, and vishing as significant cybersecurity threats hindering digital banking adoption. The study also summarized 13 preventive tools.
- Judijanto, L., Kaaffah, F.M., Muthmainah, H.N., and Vandika, A.Y. (2024) Explored computer network security's role in safeguarding financial data and customer trust in Indonesia's digitalizing financial sector. It highlights the need for collaborative efforts to address cybersecurity challenges.
- North Press (2024) This literature review explores the pivotal role of computer network security in safeguarding financial data, transactions, and customer trust within this dynamic context.
- Acharya, S. and Joshi, S. (2020) Studied the impact of cyber-attacks on banking institutions in India, focusing on safety mechanisms and preventive measures.
- Vanini et al. (2021) Emphasized that all parties involved in digital transactions must remain aware of cyber threats and update their knowledge to minimize the adverse impacts on digital banking adoption.
- Accenture (2019) Provided insights into the cost of cybercrime in the USA